Lab 11 - Perimeters and Infill - Strength vs Speed

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 03: Process Improvement for AM TSC objective: A5, A6 - apply wall/infill optimisation and assess its measured impact on time and material

Goal

Vary wall count and infill density on the same model in Cura, and record how each change moves print time, material use and expected strength.

What you'll build

A comparison table of wall/infill settings against sliced print time and material grams.

Tools: Ultimaker Cura, cube10.stl, line-width-test.stl, cube20 hole-wall variants

Steps

1. Open cube10.stl and slice with the defaults - 2 walls, 20% infill; record time and grams.

```
Slice > record: baseline
```

2. Raise the wall line count to 4 and re-slice; record the change.

```
Print Settings > Walls > Wall Line Count - 4
```

3. Reset walls to 2, then set infill density to 12% and re-slice; record.

```
Infill > Density - 12%
```

4. Set infill density to 50% and re-slice; record.

```
Infill > Density - 50%
```

5. In Preview, inspect how the infill lattice and wall thickness change at each setting.

```
Preview > layer slider
```

6. Tabulate the four runs: time, grams, and where the strength actually comes from (walls more than infill).

```
Table: setting -> time -> grams
```

7. Open line-width-test.stl and the two cube20 hole-wall variants; inspect Preview for missing thin features, then record the smallest feature and hole wall your profile resolves continuously.

```
Open three diagnostic models > Preview > Color Scheme: Line Type
```

Evidence to capture

- Wall/infill table with time, material and expected load-path notes.
- Cura Preview comparison of the line-width and matched hole-wall models.

- Measured hole diameter and wall thickness from any printed benchmark.

Test it

Your evidence separates wall-count, infill and line-width effects; the matched hole models show how wall planning changes continuity and dimensional accuracy, and you can justify settings for functional versus decorative parts.

Files in this folder

- cube10.stl - Core sample model used by the lab procedure.
- line-width-test.stl - Thin-wall benchmark for nozzle/line-width and single-wall behaviour.
- cube20-hole-0-wall.stl - Hole benchmark with the wall-control baseline geometry.
- cube20-hole-1-wall.stl - Matched hole benchmark for testing the effect of an additional wall.